

ATIVIDADE 12: SUB-REDES — PARTE 7

Faculdade com 2 laboratórios, 45 máquinas por lab. Rede Base Adotada: 192.168.20.0/24.

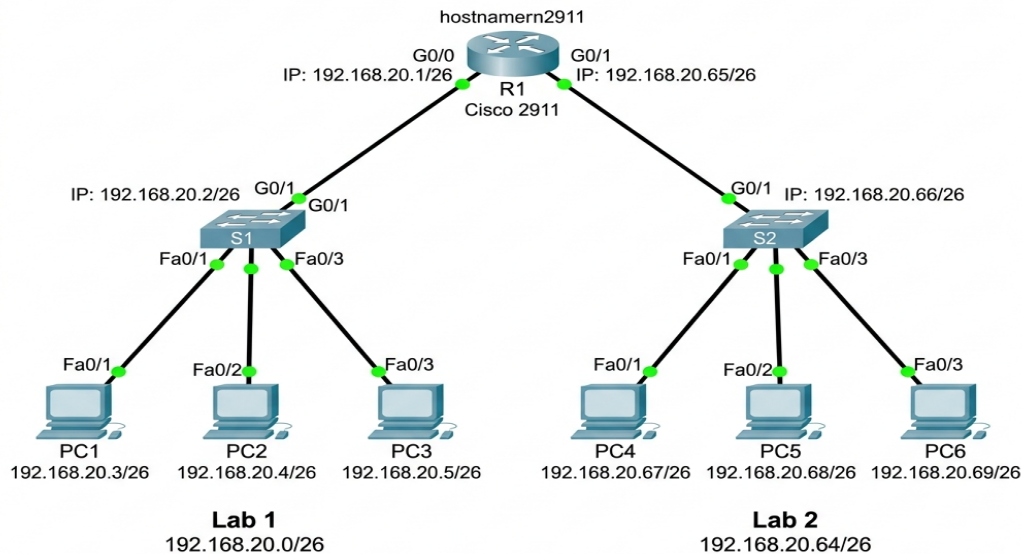
1. Memória de Cálculo

Para 45 hosts, calculamos $2^h - 2 \geq 45 \rightarrow h=6$ (62 hosts válidos). A máscara de rede passará a /26, com incrementos de bloco de 64 em 64.

Sub-rede	Endereço	Primeiro Válido	Último Válido	Broadcast
Lab 1	192.168.20.0/26	192.168.20.1	192.168.20.62	192.168.20.63
Lab 2	192.168.20.64/26	192.168.20.65	192.168.20.126	192.168.20.127

2. Topologia no Packet Tracer

Cisco Packet Tracer Network Topology: Subnets Lab 1 and Lab 2



3. Configurações de IP dos Hosts (1 por sub-rede)

Host Lab 1

IPv4 Address: 192.168.20.2
Subnet Mask: 255.255.255.192
Default Gateway: 192.168.20.1

Host Lab 2

IPv4 Address: 192.168.20.66
Subnet Mask: 255.255.255.192
Default Gateway: 192.168.20.65

4. Testes de Conectividade (Ping L3)

```
C:\>  
C:\> ping 192.168.20.66  
Pinging 192.168.20.66 with 32 bytes of data:  
Reply from 192.168.20.66: bytes=32 time<1ms TTL=128  
Reply from 192.168.20.66: bytes=32 time<1ms TTL=128  
Reply from 192.168.20.66: bytes=32 time<1ms TTL=128  
Reply from 192.168.20.66: bytes=32 time<1ms TTL=128
```